

第六章 刚体力学

§1 刚体运动学

1. 自由度

约束: $|\vec{r}_i - \vec{r}_j| = r_{ij} \quad (j < i)$

$i=1$: 无

$i=2$: 1个

$i=3$: 2个

⋮

自由度 = 0

↳ 基点: $A(x_A, y_A, z_A)$

绕转: (α, β, γ)

特殊

平移: 3个

定轴转动: 1个

平面平行运动: $\underline{2} + \underline{1} = 3$ 个
平动 转动

定点运动: 3个

2. 刚体的运动

位移定理: $\delta \vec{r} = \underbrace{\delta \vec{r}_0}_{\text{基点的运动}} + \underbrace{\delta \vec{\omega} \times \vec{r}}_{\text{绕转}}$

欧拉定理: 定点运动的刚体的任一位置移动可通过绕一定点的转动达到.

空间坐标系 $O-x'y'z'$: 固定原点.

质心坐标系 $O-xyz$: 以刚体上一点为原点.

3. 刚体内各点的运动

$$\vec{r}_P = \vec{r}_O + (\vec{r}_P - \vec{r}_O) = \vec{r}_O + \vec{r}'$$

$$\vec{v}_P = \frac{d\vec{r}_O}{dt} + \frac{d}{dt}(\vec{r}_P - \vec{r}_O) = \vec{v}_O + \vec{\omega} \times \vec{r}'$$

$$\vec{a}_P = \frac{d\vec{v}_P}{dt} = \vec{a}_O + \frac{d}{dt}(\vec{\omega} \times \vec{r}') = \vec{a}_O + \vec{\beta} \times \vec{r}' + (\vec{\omega} \times \vec{\omega} \times \vec{r}')$$



例：半径为 \$r\$ 的薄圆盘垂直地绕半径为 \$R\$ 的大圆无滑动滚动，

求 \$t=0\$ 时在圆盘最高点 \$M\$ 点的加速度

① 地球

$$\Omega R = \omega_0 r$$

\$\varphi, \theta\$

$$\vec{\omega} = [-\omega_0 \cos \varphi \hat{i} + (-\omega_0 \sin \varphi \hat{j} + \Omega \hat{k}]$$

$$= -\frac{R}{r} \cos \varphi \Omega \hat{i} + (-\frac{R}{r} \sin \varphi \Omega \hat{j}) + \Omega \hat{k}$$

$$\vec{\beta} = \frac{R}{r} \sin \varphi \Omega^2 \hat{i} - \frac{R}{r} \cos \varphi \Omega^2 \hat{j}$$

$$\text{取 } t=0 \text{ 时 } \varphi=0 \rightarrow \varphi=\Omega t \quad \theta = \frac{1}{r} \varphi = \frac{1}{r} \Omega t$$

$$\vec{r}_M = [R \cos \varphi - r \sin \theta \sin \varphi] \hat{i} + [R \sin \varphi + r \sin \theta \cos \varphi] \hat{j} + r \cos \theta \hat{k}$$

$$\vec{a}_M = \vec{\beta} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$= [\frac{R}{r} \sin(\omega t) \Omega^2 \cdot [R \sin \varphi + r \sin \theta \cos \varphi] \hat{k}$$

$$- \frac{R}{r} \sin \varphi \cdot \Omega \cdot r \cos \theta \hat{j} + \frac{R}{r} \cos \varphi \Omega^2 \cdot (R \cos \varphi - r \sin \theta \sin \varphi) \hat{i}$$

$$- \frac{R}{r} \cos \varphi \Omega^2 r \cos \theta \hat{i}] +$$

$$[-\frac{R}{r} \cos \varphi \Omega \hat{i} + (-\frac{R}{r} \sin \varphi \Omega \hat{j}) + \Omega \hat{k}] \times$$

$$[-\frac{R}{r} \cos \varphi \Omega \hat{i} + (-\frac{R}{r} \sin \varphi \Omega \hat{j}) + \Omega \hat{k}] \times$$

$$[[R \cos \varphi - r \sin \theta \sin \varphi] \hat{i} + [R \sin \varphi + r \sin \theta \cos \varphi] \hat{j} + r \cos \theta \hat{k}]$$

取 $t=0$ 即 $\varphi = \theta = 0$

$$\begin{aligned}\vec{a}_M &= \frac{R^2}{r} \Omega^2 \hat{k} - R \Omega^2 \hat{i} + \\ &\quad \left(-\frac{R}{r} \Omega \hat{i} + \Omega \hat{k} \right) \times \left[\left(-\frac{R}{r} \Omega \hat{i} + \Omega \hat{k} \right) \times (R \hat{i} + r \hat{k}) \right] \\ &= \frac{R^2}{r} \Omega^2 \hat{k} - R \Omega^2 \hat{i} + \left(-\frac{R}{r} \Omega \hat{i} + \Omega \hat{k} \right) \times 2R \Omega \hat{j} \\ &= -3R \Omega^2 \hat{i} - \frac{R^2}{r} \Omega^2 \hat{k}\end{aligned}$$

② 随圆盘转动系

4. 基点选取

$$\vec{v}_B = \vec{v}_A + \vec{\omega} \times (\vec{r}_B - \vec{r}_A)$$

$$\begin{aligned} \vec{v}_P &= \vec{v}_B + \vec{\omega} \times (\vec{r}_P - \vec{r}_B) = \vec{v}_B - \vec{\omega} \times (\vec{r}_B - \vec{r}_A) + \vec{\omega} \times (\vec{r}_P - \vec{r}_A) \\ &= \vec{v}_A + \vec{\omega} \times (\vec{r}_P - \vec{r}_A) \end{aligned}$$

$\Rightarrow \omega$ 与选取基点无关

5. 角速度矢量

$$\vec{v}_P = \vec{v}_A + \vec{\omega}_1 \times (\vec{r}_P - \vec{r}_A) + \vec{\omega}_2 \times (\vec{r}_P - \vec{r}_A)$$

$$= \vec{v}_A + (\vec{\omega}_1 + \vec{\omega}_2) \times (\vec{r}_P - \vec{r}_A) = \vec{v}_A + \vec{\omega} \times (\vec{r}_P - \vec{r}_A) \quad \phi \hat{n}$$

$\Rightarrow \omega$ 可叠加

$$b. \quad \vec{r}_\perp = (\vec{r} \cdot \hat{n}) \cdot \hat{n}$$

$$\vec{r}_\parallel = \vec{r} - \vec{r}_\perp = \vec{r} - (\vec{r} \cdot \hat{n}) \cdot \hat{n}$$

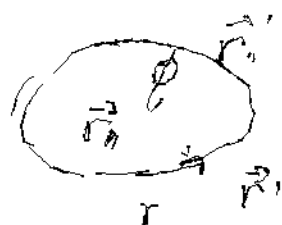
$$\vec{r}'_\perp = \vec{r}_\parallel \cos \phi + (\hat{n} \times \vec{r}_\parallel) \sin \phi$$

$$= [\vec{r} - (\vec{r} \cdot \hat{n}) \hat{n}] \cos \phi + (\hat{n} \times \vec{r}) \sin \phi$$

$$\vec{r}'_\parallel = (\vec{r}' \cdot \hat{n}) \cdot \hat{n} = \vec{r}_\parallel$$

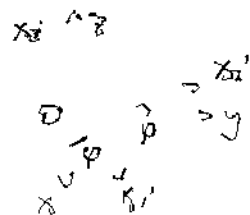
$$\Rightarrow \vec{r}' = \vec{r}'_\parallel + \vec{r}'_\perp$$

$$= (\vec{r} \cdot \hat{n}) \cdot \hat{n} + [\vec{r} - (\vec{r} \cdot \hat{n}) \hat{n}] \cos \phi + (\hat{n} \times \vec{r}) \sin \phi$$



7. 欧拉角

空间坐标系 $Oxyz$
 转动坐标系 $Ox_1x_2x_3$



① 进动角 $\theta \in [0, \pi)$

→ 绕 z 轴转角

② 章动角 $\theta \in [0, \pi)$?

→ 绕 Ox_1 轴转角

③ 自旋角 $\psi \in [0, 2\pi)$

→ 绕 Ox_3 轴转角

$$\vec{\omega} = \dot{\psi} \vec{k} + \dot{\theta} \vec{n} + \dot{\phi} \vec{k}'$$

其中 $\vec{k}, \vec{n}, \vec{k}'$ 分别为 Oz, Ox_2, Ox_3 轴单位向量

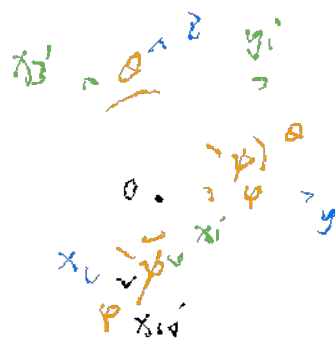
$$\vec{k} = \sin \theta \sin \psi \vec{i}' + \sin \theta \cos \psi \vec{j}' + \cos \theta \vec{k}'$$

$$\vec{n} = \cos \psi \vec{i}' - \sin \psi \vec{j}'$$

$$\vec{\omega} = (\dot{\psi} \sin \theta \sin \psi + \dot{\theta} \cos \psi) \vec{i}'$$

$$+ (\dot{\psi} \sin \theta \cos \psi - \dot{\theta} \sin \psi) \vec{j}'$$

$$+ (\dot{\psi} \cos \theta + \dot{\phi}) \vec{k}'$$



§ 2. 刚体动力学

1. 运动定理

动量 $\times 3$, 角动量 $\times 3$, 动能 $\times 1$

刚体自由度 $i=6$

$\rightarrow$ 6个方程独立

$$\left\{ \begin{array}{l} \text{动量定理: } m\vec{v} = \vec{P} = \sum_i \vec{F}_i \quad \text{主矢 (诸力作用线不交于一点)} \\ \text{角动量定理: } \vec{L} = \vec{M} = \sum_i \vec{M}_i \quad \text{主矩} \end{array} \right.$$

$$\text{动能定理: } T_2 - T_1 = W_{\text{内}} + W_{\text{外}}$$

$$\begin{aligned} W_{\text{内}} &= \sum_{ij} (\vec{F}_{ij} \cdot d\vec{r}_i + \vec{F}_{ji} \cdot d\vec{r}_j) \\ &= \sum_{ij} \vec{F}_{ij} (d\vec{r}_i - d\vec{r}_j) = \sum_{ij} \vec{F}_{ij} \cdot d\vec{r}_{ij} \\ &= \sum_{ij} |\vec{F}_{ij}| \cdot r_{ij} \cdot d\vec{r}_{ij} \end{aligned}$$

$$\text{又 } r_{ij}^2 = \text{const}$$

$$d(r_{ij}^2) = 2\vec{r}_{ij} \cdot d\vec{r}_{ij} = 2\vec{r}_{ij} \cdot \underbrace{d\vec{r}_{ij}}_{=0} = 0$$

$$\hookrightarrow W_{\text{内}} = 0$$

$$T_2 - T_1 = W_{\text{外}}$$

2. 角动量, 动能

$$\begin{aligned} (1) \quad L &= \sum_i \vec{r}_i \times (m_i \vec{v}_i) = \sum_i m_i \cdot \vec{r}_i \times (\vec{\omega} \times \vec{r}_i) \\ &= \sum_i m_i [\vec{\omega} (\vec{r}_i \cdot \vec{r}_i) - \vec{r}_i (\vec{r}_i \cdot \vec{\omega})] \\ &= \sum_i m_i [(x_i^2 + y_i^2 + z_i^2) (\omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}) \end{aligned}$$

$$= (x_i \omega_x + y_i \omega_y + z_i \omega_z) (x_i \hat{i} + y_i \hat{j} + z_i \hat{k}) \\ = L_x \cdot \hat{i} + L_y \cdot \hat{j} + L_z \cdot \hat{k}$$

$$\Rightarrow L_x = m_i [(x_i^2 + y_i^2 + z_i^2) \omega_x - x_i^2 \omega_x - x_i y_i \omega_y - x_i z_i \omega_z] \\ = m_i [(y_i^2 + z_i^2) \omega_x - x_i y_i \omega_y - x_i z_i \omega_z]$$

$$L_y = m_i [-x_i y_i \omega_x + (x_i^2 + z_i^2) \omega_y - y_i z_i \omega_z]$$

$$L_z = m_i [-x_i z_i \omega_x - y_i z_i \omega_y + (x_i^2 + y_i^2) \omega_z]$$

写为矩阵形式

$$= \begin{pmatrix} I_{11} & -I_{12} & -I_{13} \\ -I_{21} & I_{22} & -I_{23} \\ -I_{31} & -I_{32} & I_{33} \end{pmatrix}$$

$$\vec{L} = I \vec{\omega} \quad \text{其中 } I = \begin{pmatrix} I_{11} & -I_{12} & -I_{13} \\ -I_{21} & I_{22} & -I_{23} \\ -I_{31} & -I_{32} & I_{33} \end{pmatrix}$$

转动惯量

$$\begin{cases} x: I_{11} = I_{xx} = \sum_i m_i (y_i^2 + z_i^2) \\ y: I_{22} = I_{yy} = \sum_i m_i (x_i^2 + z_i^2) \\ z: I_{33} = I_{zz} = \sum_i m_i (x_i^2 + y_i^2) \end{cases}$$

惯量积

$$\begin{cases} I_{12} = I_{21} = I_{xy} = I_{yx} = \sum_i m_i x_i y_i \\ I_{13} = I_{31} = I_{xz} = I_{zx} = \sum_i m_i x_i z_i \\ I_{23} = I_{32} = I_{yz} = I_{zy} = \sum_i m_i y_i z_i \end{cases}$$

基点 O → 一般运动

$$\vec{r}_i = \underbrace{\vec{r}_0}_{\text{相对点}} + \underbrace{\vec{r}_i'}_{\text{相对点}}, \quad \vec{v}_i = \vec{v}_0 + \vec{\omega} \times \vec{r}_i'$$

$$\vec{L} = \sum_i m_i \vec{r}_i \times \vec{v}_i = \sum_i m_i (\vec{r}_0 + \vec{r}_i') \times (\vec{v}_0 + \vec{\omega} \times \vec{r}_i')$$

$$= \sum_i m_i \vec{r}_0 \times \vec{v}_0 + \sum_i m_i \vec{r}_0 \times \vec{v}_i + \sum_i m_i \vec{r}_0 \times (\vec{\omega} \times \vec{r}_i') + \sum_i m_i \vec{r}_i' \times (\vec{\omega} \times \vec{r}_i')$$

若以质心 C 为坐标原点, $\sum_i m_i \vec{r}_0 = 0$, $\sum_i m_i (\vec{\omega} \times \vec{r}_i') = \sum_i m_i \vec{v}_i = 0$

$$\vec{L} = m \vec{r}_i \times \vec{V}_i + I_C \cdot \vec{\omega} = \vec{L}_C + \vec{L}_r$$

质心系角动量定理: 角动量 = 质心角动量 + 相对质心角动量

$$(2) \quad T = \sum_i \frac{1}{2} m_i V_i^2$$

定点转动时:

$$T = \sum_i \frac{1}{2} m_i (\vec{\omega} \times \vec{r}_i)^2$$

$$= \sum_i \frac{1}{2} m_i \vec{V}_i \cdot (\vec{\omega} \times \vec{r}_i)$$

很麻烦
轮换

$$= \sum_i \frac{1}{2} \vec{\omega} \cdot (\vec{r}_i \times m_i \vec{V}_i)$$

$$= \frac{1}{2} \vec{\omega} \cdot \vec{L}$$

矩阵
写法

$$\hookrightarrow = \frac{1}{2} \omega^T I \omega$$

一般运动时:

$$T = \sum_i \frac{1}{2} m_i V_i^2$$

$$= \sum_i \frac{1}{2} m_i (\vec{V}_0 + \vec{\omega} \times \vec{r}_i')^2$$

$$= \sum_i \frac{1}{2} m_i \vec{V}_0^2 + \sum_i m_i \vec{V}_0 \cdot (\vec{\omega} \times \vec{r}_i') + \sum_i \frac{1}{2} m_i (\vec{\omega} \times \vec{r}_i')^2$$

$$= \sum_i \frac{1}{2} m_i \vec{V}_0^2 + \sum_i m_i \vec{r}_i' \cdot (\vec{V}_0 \times \vec{\omega}) + \sum_i \frac{1}{2} m_i V_i'^2$$

选取质心 C 为坐标原点 $\sum_i m_i \vec{r}_i' = 0$

$$T = \sum_i \frac{1}{2} m_i V_0^2 + \sum_i \frac{1}{2} m_i V_{cr}^2 = T_C + T_{cr}$$

3. 惯量张量 惯量椭球

$$I = \begin{pmatrix} I_{11} & -I_{12} & -I_{13} \\ -I_{11} & I_{22} & -I_{23} \\ -I_{13} & -I_{23} & I_{33} \end{pmatrix}$$

对称化

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

$$L_x = I_1 \omega_x, \quad L_y = I_2 \omega_y, \quad L_z = I_3 \omega_z$$

主轴坐标系 / 惯量主轴

I_1, I_2, I_3 : 主转动惯量

$$T = \frac{1}{2} (I_1 \omega_x^2 + I_2 \omega_y^2 + I_3 \omega_z^2)$$

$$\begin{aligned} \vec{\omega} = & (\dot{\psi} \sin \theta \sin \psi + \dot{\theta} \cos \psi) \hat{z}' \\ & + (\dot{\psi} \sin \theta \cos \psi - \dot{\theta} \sin \psi) \hat{y}' \\ & + (\dot{\psi} \cos \theta + \dot{\phi}) \hat{K} \end{aligned}$$

$$\Rightarrow L_x = I_1 \cdot (\dot{\psi} \sin \theta \sin \psi + \dot{\theta} \cos \psi)$$

$$L_y = I_2 \cdot (\dot{\psi} \sin \theta \cos \psi - \dot{\theta} \sin \psi)$$

$$L_z = I_3 \cdot (\dot{\psi} \cos \theta + \dot{\phi})$$

$$\begin{aligned} T = & \frac{1}{2} [I_1 (\dot{\psi} \sin \theta \sin \psi + \dot{\theta} \cos \psi)^2 + I_2 (\dot{\psi} \sin \theta \cos \psi - \dot{\theta} \sin \psi)^2 \\ & + I_3 (\dot{\psi} \cos \theta + \dot{\phi})^2] \end{aligned}$$

对称刚体: $I_1 = I_2$

$$\hookrightarrow T = \frac{1}{2} I_1 (\dot{\psi}^2 \sin^2 \theta + \dot{\theta}^2) + \frac{1}{2} I_3 (\dot{\psi} \cos \theta + \dot{\phi})^2$$

例: 计算匀质正立方体的惯量主轴方向和主转动惯量

先计算惯量张量:

$$\begin{aligned} I_{11} = I_{22} = I_{33} &= \int (y^2 + z^2) dm = \iiint \rho (y^2 + z^2) dx dy dz \\ &= \rho a \cdot \iint (y^2 + z^2) dy dz \\ &= \rho a \cdot \left(\frac{1}{3} a^3 + \frac{1}{3} a^3 \right) = \frac{2}{3} m a^2 \end{aligned}$$

$$I_{12} = \int xy \, dm = \iiint xy \rho \cdot dx dy dz = \rho \cdot \left(\frac{1}{2}a^2\right) \cdot \left(\frac{1}{2}a^2\right) \cdot a = \frac{1}{4}ma^2$$

$$= I_{21} = I_{13} = I_{31} = I_{23} = I_{32}$$

$$I = \begin{bmatrix} \frac{2}{3}ma^2 - \frac{1}{4}ma^2 & -\frac{1}{4}ma^2 \\ -\frac{1}{4}ma^2 & \frac{2}{3}ma^2 - \frac{1}{4}ma^2 \\ -\frac{1}{4}ma^2 & -\frac{1}{4}ma^2 & \frac{2}{3}ma^2 \end{bmatrix}$$

角动矢量:

$$I\omega =$$

惯量椭球：

取过O点，方向余弦为 α, β, γ 的直线 $l: \vec{r} = \alpha \vec{i} + \beta \vec{j} + \gamma \vec{k}$

求刚体对此直线转动惯量 I

$$I = \sum_i m_i d_i^2 = \sum_i m_i [\vec{r}_i - (\vec{r}_i \cdot \hat{n}) \cdot \hat{n}]^2$$

$$= \sum_i m_i [\vec{r}_i^2 - 2\vec{r}_i \cdot \hat{n} (\vec{r}_i \cdot \hat{n}) + (\vec{r}_i \cdot \hat{n})^2 \hat{n}^2]$$

$$= \sum_i m_i [\vec{r}_i^2 - (\vec{r}_i \cdot \hat{n})^2]$$

$$= \sum_i m_i [x_i^2 + y_i^2 + z_i^2 - (x_i \alpha + y_i \beta + z_i \gamma)^2]$$

$$= \sum_i m_i [x_i^2 (1 - \alpha^2) + y_i^2 (1 - \beta^2) + z_i^2 (1 - \gamma^2)$$

$$- 2x_i y_i \alpha \beta - 2y_i z_i \beta \gamma - 2x_i z_i \alpha \gamma]$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1$$

$$= \sum_i m_i [x_i^2 (\beta^2 + \gamma^2) + y_i^2 (\alpha^2 + \gamma^2) + z_i^2 (\alpha^2 + \beta^2)$$

$$- 2x_i y_i \alpha \beta - 2y_i z_i \beta \gamma - 2x_i z_i \alpha \gamma]$$

$$= \alpha^2 I_{11} + \beta^2 I_{22} + \gamma^2 I_{33} - 2\alpha\beta I_{12} - 2\alpha\gamma I_{13} - 2\beta\gamma I_{23}$$

$$\text{取 } l \text{ 上一点 } P(\frac{\alpha}{r_1}, \frac{\beta}{r_2}, \frac{\gamma}{r_3}) \quad (P\vec{O} = \frac{1}{r_1}) \quad I_{Op} = \frac{1}{\sigma_P^2}$$

$$I = x_P^2 I_{11} + y_P^2 I_{22} + z_P^2 I_{33} - 2x_P y_P I_{12} - 2x_P z_P I_{13} - 2y_P z_P I_{23}$$

$= F(x, y, z)$ 为一椭球面，称为惯量椭球

惯量主轴:

$$\text{设主轴 } \vec{n} = \alpha \vec{i} + \beta \vec{j} + \gamma \vec{k}$$

$$\text{角动量 } \vec{L} = I \omega \vec{n} = I \vec{\omega} = \vec{I} \vec{\omega}$$

$\Rightarrow$ 主轴 $\vec{n}$ 是 $\vec{I}$ 的本征矢量,

$$|I - \lambda E| = 0, \text{ 解出 } \lambda \text{ 和对应 } \vec{n}$$

$$\text{满足 } I = P^T \text{diag}(I_1, I_2, I_3) P$$

其中 I_1, I_2, I_3 为刚体沿惯量主轴的转动惯量, 称主转动惯量.

$$\text{主轴坐标下: } I_1 x^2 + I_2 y^2 + I_3 z^2 = 1$$

$$\omega = \alpha \omega \vec{i} + \beta \omega \vec{j} + \gamma \omega \vec{k} \text{ 在 } F(x, y, z) \text{ 上}$$

$$\nabla F|_P = (2I_1 x_P - 2I_2 y_P - 2I_3 z_P) \vec{i} + (2I_2 y_P - 2I_1 x_P - 2I_3 z_P) \vec{j} \\ + (2I_3 z_P - 2I_1 x_P - 2I_2 y_P) \vec{k}$$

$$= \frac{2}{\sqrt{2} \cdot \omega} [(I_1 \alpha \omega - I_2 \beta \omega - I_3 \gamma \omega) \vec{i} + (I_2 \beta \omega - I_1 \alpha \omega - I_3 \gamma \omega) \vec{j} \\ + (I_3 \gamma \omega - I_1 \alpha \omega - I_2 \beta \omega) \vec{k}]$$

$$= \frac{2}{\omega \sqrt{2}} I \vec{\omega} = \frac{2}{\omega \sqrt{2}} \cdot \vec{L}$$



4. 欧拉动力学方程 质心位移

$$\delta \vec{r}_i = \underline{\delta \vec{r}_0} + \vec{n} \delta \varphi \times (\vec{r}_i - \vec{r}_0)$$

达朗贝尔原理:

$$\begin{aligned} & \sum (\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_i = 0 \\ & = \sum (\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot [\delta \vec{r}_0 + \vec{n} \delta \varphi \times (\vec{r}_i - \vec{r}_0)] \\ & = \sum \left[\vec{F}_i \cdot \delta \vec{r}_0 - m_i \ddot{\vec{r}}_i \cdot \delta \vec{r}_0 + \vec{F}_i \cdot \vec{n} \delta \varphi \times (\vec{r}_i - \vec{r}_0) - m_i \ddot{\vec{r}}_i \cdot \vec{n} \delta \varphi \times (\vec{r}_i - \vec{r}_0) \right] \\ & = \sum \left[(\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot \delta \vec{r}_0 + [(\vec{r}_i - \vec{r}_0) \times \vec{F}_i - (\vec{r}_i - \vec{r}_0) \times (m_i \ddot{\vec{r}}_i)] \cdot \vec{n} \delta \varphi \right] \end{aligned}$$

$$\begin{aligned} L &= \sum (\vec{r}_i - \vec{r}_0) \times m_i \dot{\vec{r}}_i, \quad \dot{L} = \sum \left[\vec{r}_i \times m_i \dot{\vec{r}}_i + (\vec{r}_i - \vec{r}_0) \times m_i \dot{\vec{r}}_i \right] \\ &= (\vec{F} - m \ddot{\vec{r}}_0) \cdot \delta \vec{r}_0 + (\vec{L} - \dot{L}) \cdot \vec{n} \delta \varphi = 0 \end{aligned}$$

$\Rightarrow$ 质心运动定理: $\vec{F} = m \ddot{\vec{r}}_0$

质心角转动定理: $\vec{L} = \dot{L}$

$$\vec{L} = \dot{L}$$

$$\begin{cases} \frac{d}{dt} (I_{11} \omega_x - I_{12} \omega_y - I_{13} \omega_z) = M_x \\ \frac{d}{dt} (-I_{21} \omega_x + I_{22} \omega_y - I_{23} \omega_z) = M_y \\ \frac{d}{dt} (-I_{31} \omega_x - I_{32} \omega_y + I_{33} \omega_z) = M_z \end{cases}$$

$$\frac{d}{dt} \vec{L} = \dot{\vec{L}} + \underline{\vec{\omega} \times \vec{L}}$$

绝对变化率 相对变化率 牵连变化率

在任意坐标系中, $\vec{L} = (L_1, L_2, L_3)$

$$\begin{cases} L_1 = I_{11} \dot{\omega}_1 - I_{12} \dot{\omega}_2 - I_{13} \dot{\omega}_3 \\ L_2 = -I_{21} \dot{\omega}_1 + I_{22} \dot{\omega}_2 - I_{23} \dot{\omega}_3 \\ L_3 = -I_{31} \dot{\omega}_1 - I_{32} \dot{\omega}_2 + I_{33} \dot{\omega}_3 \end{cases}$$

刚体定点运动的欧拉动力学方程:

$$\dot{\vec{L}} = \vec{M} + \vec{L} \times \vec{\omega} \quad (2 \times 1)$$

$$\begin{cases} I_{11} \dot{\omega}_1 - I_{12} \dot{\omega}_2 - I_{13} \dot{\omega}_3 = M_1 + I_{13} \omega_1 \omega_3 - I_{12} \omega_1 \omega_2 + (I_{22} - I_{33}) \omega_2 \omega_3 + I_{23} (\omega_2^2 - \omega_3^2) \\ -I_{21} \dot{\omega}_1 - I_{22} \dot{\omega}_2 - I_{23} \dot{\omega}_3 = M_2 + I_{12} \omega_1 \omega_2 - I_{23} \omega_1 \omega_3 + (I_{33} - I_{11}) \omega_3 \omega_1 + I_{31} (\omega_3^2 - \omega_1^2) \\ -I_{31} \dot{\omega}_1 - I_{32} \dot{\omega}_2 + I_{33} \dot{\omega}_3 = M_3 + I_{23} \omega_3 \omega_1 - I_{13} \omega_3 \omega_2 + (I_{11} - I_{22}) \omega_1 \omega_2 + I_{12} (\omega_1^2 - \omega_2^2) \end{cases}$$

取为体坐标系的动量主轴. (惯量积为0)

$$\begin{cases} I_{11} \dot{\omega}_1 = M_1 + (I_{22} - I_{33}) \omega_2 \omega_3 \\ I_{22} \dot{\omega}_2 = M_2 + (I_{33} - I_{11}) \omega_3 \omega_1 \\ I_{33} \dot{\omega}_3 = M_3 + (I_{11} - I_{22}) \omega_1 \omega_2 \end{cases}$$

★ 取刚体静止系

$\dot{\vec{L}} = 0$ - $\vec{L}$ 中包含惯性力矩:

$$\begin{aligned} \dot{\vec{L}} = 0 &= \vec{M} + \int \vec{r} \times [-\vec{\omega} \times (\vec{\omega} \times \vec{r})] dm + \int \vec{r} \times \left[\frac{d\vec{\omega}}{dt} \times \vec{r} \right] dm \\ &= \vec{M} - \int \vec{r} \times [\vec{\omega} \times (\vec{\omega} \times \vec{r})] dm - \int \vec{r} \times \left[\frac{d\vec{\omega}}{dt} \times \vec{r} \right] dm \\ &= \vec{M} + \left[\int \vec{r} \times (\vec{\omega} \times \vec{r}) dm \right] \times \vec{\omega} - \frac{d}{dt} \int \vec{r} \times (\vec{\omega} \times \vec{r}) dm \\ &= \vec{M} + \vec{L} \times \vec{\omega} - \frac{d}{dt} (\vec{L}) \quad \text{也所以得到欧拉动力学方程} \end{aligned}$$

拉格朗日方程:

$$L = T - U$$

$$= \frac{1}{2} \left[I_1 \dot{\theta}^2 + I_2 (\dot{\theta} \cos \psi + \dot{\psi} \sin \theta \sin \psi)^2 + I_3 (\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta) + I_3 (\dot{\psi} + \dot{\theta} \cos \theta)^2 \right] - U$$

动能定理:

$$\begin{aligned} dT &= \sum_i \vec{F}_i \cdot d\vec{r}_i \\ &= \sum_i \vec{F}_i \cdot \vec{n} \delta \varphi \times \vec{r}_i \\ &= \sum_i \vec{r}_i \times \vec{F}_i \cdot \vec{\omega} dt = \sum_i M_i \cdot \vec{\omega} \cdot dt \end{aligned}$$

定点运动的刚体动能改变量 = 外力矩做功

刚体作用力:

$$M_1 = (I_2 - I_3) \omega_2 \omega_3 - I_{12} \dot{\omega}_1$$

$$M_2 = (I_3 - I_1) \omega_3 \omega_1 - I_{21} \dot{\omega}_2$$

$$L_3 = (I_1 - I_2) \omega_1 \omega_2 - I_3 \omega_3$$

$$L = \frac{1}{2} m \dot{r}^2 + m \dot{\theta}^2 + m \dot{\phi}^2$$

§ 6.3 刚体的平移 定轴转动 平面平行运动

例: 17 0.18

$$(R-r) \dot{\theta} = r \omega = v$$

$$m g (R-r) = \frac{1}{2} m (R-r)^2 \dot{\theta}^2 + \frac{1}{2} \cdot \frac{2}{3} m r^2 \omega^2$$

$$\hookrightarrow v = \sqrt{\frac{10}{7} g (R-r)}$$

17 0.20 例 3

$$L = \frac{1}{2} \cdot \frac{3}{5} m \dot{x}_C^2 + \frac{1}{2} m (\dot{x}_C^2 + \dot{y}_C^2) + \frac{1}{2} \cdot \frac{1}{12} m (2R \sin \alpha)^2 \dot{\theta}^2 - m g y_C$$

$$\begin{cases} x_C = R \cos \alpha \sin \theta & \dot{x}_C = R \cos \alpha \cos \theta \dot{\theta} \\ y_C = R \cos \alpha (1 - \cos \theta) & \dot{y}_C = R \cos \alpha \sin \theta \dot{\theta} \end{cases}$$

$$L = \frac{3}{10} m \dot{x}_C^2 + \frac{1}{2} m R^2 \cos^2 \alpha \dot{\theta}^2 + \frac{1}{6} m R^2 \sin^2 \alpha \dot{\theta}^2 - m g R \cos \alpha (1 - \cos \theta)$$

$$\text{运动方程: } m \ddot{x}_C + m R \cos \alpha \cos \theta \ddot{\theta} = 0$$

$$L = \frac{3}{10} \frac{m^2}{m} R^2 \cos^2 \alpha \cos^2 \theta \dot{\theta}^2 + \frac{1}{2} m R^2 \cos^2 \alpha \dot{\theta}^2 + \frac{1}{6} m R^2 \sin^2 \alpha \dot{\theta}^2 - m g R \cos \alpha (1 - \cos \theta)$$

$$\frac{\partial L}{\partial \theta} = -\frac{3}{5} \frac{m^2}{m} R^2 \cos^2 \alpha \cos \theta \sin \theta \dot{\theta}^2 - m g R \cos \alpha \sin \theta$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{3}{5} \frac{m^2}{m} R^2 \cos^2 \alpha \frac{d}{dt} (\cos^2 \theta \cdot \dot{\theta}) + m R^2 \cos^2 \alpha \ddot{\theta} + \frac{1}{3} m R^2 \sin^2 \alpha \ddot{\theta}$$

$$= \frac{3}{5} \frac{m^2}{m} R^2 \cos^2 \alpha (\cos^2 \theta \ddot{\theta} - 2 \cos \theta \sin \theta \cdot \dot{\theta}^2) + m R^2 \cos^2 \alpha \ddot{\theta} + \frac{1}{3} m R^2 \sin^2 \alpha \ddot{\theta}$$

$$\Rightarrow \frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = -\frac{3}{5} \frac{m^2}{m} R^2 \cos^2 \alpha \cos \theta \sin \theta \dot{\theta}^2 - m g R \cos \alpha \sin \theta$$

$$+ \left(\frac{3}{5} \frac{m^2}{m} R^2 \cos^2 \alpha \cos^2 \theta + m R^2 \cos^2 \alpha + \frac{1}{3} m R^2 \sin^2 \alpha \right) \ddot{\theta} = 0$$

$\hookrightarrow$

§ 6.4 无外力矩的定点运动 (欧拉 - 潘索)

1. 对称刚体

$$I_1 = I_2 \rightarrow \begin{cases} I_1 \dot{\omega}_1 = (I_1 - I_3) \omega_2 \omega_3 \\ I_1 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 \\ I_3 \dot{\omega}_3 = (I_1 - I_3) \omega_1 \omega_2 = 0 \end{cases} \rightarrow \omega_3 = \text{const}$$

(3 为对称轴)

$$I_1 \ddot{\omega}_1 = (I_1 - I_3) \omega_3 \dot{\omega}_2$$

$$= (I_1 - I_3) \frac{I_3 - I_1}{I_1} \omega_3^2 \omega_1$$

$$\Rightarrow \omega_1 = A \cos\left(\frac{I_3 - I_1}{I_1} \omega_3 t + \alpha\right)$$

$$\omega_2 = A \sin\left(\frac{I_3 - I_1}{I_1} \omega_3 t + \alpha\right)$$

本体极锥：在本体坐标系中， $\vec{\omega}$ 绕对称轴 z 旋转，形成本体极锥

$$|\vec{\omega}| = \text{const} \quad \gamma = \frac{I_3 - I_1}{I_3 + I_1} \omega_3$$

$$\begin{cases} \dot{\theta} = \omega_1 \cos \psi - \omega_2 \sin \psi & = A \cos\left(\frac{I_3 - I_1}{I_1} \omega_3 t + \alpha + \psi\right) \\ \dot{\varphi} = \omega_1 \csc \theta \sin \psi + \omega_2 \csc \theta \cos \psi & = \csc \theta A \sin\left(\frac{I_3 - I_1}{I_1} \omega_3 t + \alpha + \psi\right) \\ \dot{\psi} = -\omega_1 \cot \theta \sin \psi - \omega_2 \cot \theta \cos \psi + \omega_3 & = -\cot \theta A \sin\left(\frac{I_3 - I_1}{I_1} \omega_3 t + \alpha + \psi\right) + \omega_3 \end{cases}$$

取 z 方向为 z 轴， θ 为 z 、 3 夹角， $\dot{\theta} = 0$

$$\psi = \frac{I_1 - I_3}{I_1} \omega_3 t, \quad \alpha = \frac{\pi}{2}$$

$$\dot{\psi} = \frac{I_1 - I_3}{I_1} \omega_3, \quad \dot{\theta} = \frac{I_3 \omega_3}{I_1 \cos \theta}$$

→ 规则进动

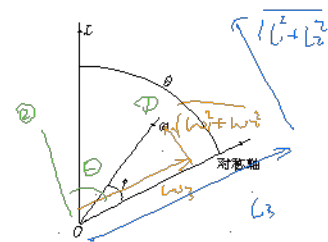
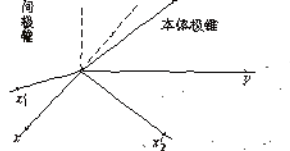
空间极锥：在空间坐标系中， $\vec{\omega}$ 绕 z 旋转，形成空间极锥。

$$\tan \theta = \frac{\sqrt{L_1^2 + L_2^2}}{L_3} = \frac{I_1 A}{I_3 \omega_3} \quad \tan \delta = \frac{\omega_1^2 + \omega_2^2}{\omega_3} = \frac{A}{\omega_3}$$



$$① I_1 > I_3, \quad \theta > 0$$

$$② I_1 < I_3, \quad \theta < 0$$



* 拉格朗日力学

$$L = \frac{1}{2} I_1 (\dot{\varphi}^2 \sin^2 \theta + \dot{\theta}^2) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\varphi} \cos \theta)^2$$

$$\frac{\partial L}{\partial \varphi} = I_1 \dot{\varphi} \sin^2 \theta + I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \cos \theta = p_{\varphi} = L_z \quad (\varphi \text{ 是沿 } z \text{ 轴的转角})$$

$$\frac{\partial L}{\partial \psi} = I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) = p_{\psi} = L_3 \quad (\psi \text{ 是沿 } x_3 \text{ 轴的转角})$$

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = I_1 \dot{\varphi} \sin \theta \cos \theta - I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \sin \theta + 2 I_1 \dot{\theta} = 0$$

* 选取 $\theta = \theta_0, \quad \dot{\theta} = 0$ 的稳定坐标系

$$L_z = I_1 \dot{\varphi} \sin^2 \theta + L_3 \cos \theta$$

$$L_3 = I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \Rightarrow \dot{\varphi} = \frac{L_3}{I_3 \cos \theta} \quad \dot{\psi} = \frac{L_3}{I_3} - \dot{\varphi} \cos \theta = L_3 \left(\frac{1}{I_3} - \frac{1}{I_1} \right)$$

$$\left[I_1 \dot{\varphi} \cos \theta - L_3 \right] \sin \theta = 0 \Rightarrow \frac{I_3 \omega_3}{I_1 \cos \theta} = \frac{I_1 - I_3}{I_1} \omega_3$$

例：作 0.22 对称陀螺作无外力矩的定点运动

试证：(1) 已在本体坐标系中以 Ω 的角速度运动

$$\text{上述运动的解为：} \quad \dot{\theta} = 0, \quad \dot{\varphi} = \frac{I_3}{I_1 \cos \theta} \omega_3, \quad \dot{\psi} = \frac{I_1 - I_3}{I_1} \omega_3$$

在本体坐标系中，

$$\vec{C} = I_1 \omega_1 \vec{i}' + I_2 \omega_2 \vec{j}' + I_3 \omega_3 \vec{k}'$$

$$= I_1 (\dot{\varphi} \sin \theta \sin \psi + \dot{\theta} \cos \psi) \vec{i}' + I_1 (\dot{\varphi} \sin \theta \cos \psi - \dot{\theta} \sin \psi) \vec{j}' + I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \vec{k}'$$

$$= I_1 \cdot \frac{I_3}{I_1 \cos \theta} \omega_3 \sin \theta \sin \left(\frac{I_1 - I_3}{I_1} \omega_3 t + \psi_0 \right) \vec{i}' + I_1 \cdot \frac{I_3}{I_1 \cos \theta} \omega_3 \sin \theta \cos \left(\frac{I_1 - I_3}{I_1} \omega_3 t + \psi_0 \right) \vec{j}'$$

$$+ I_3 \left(\frac{I_3}{I_1 \cos \theta} \omega_3 \cos \theta + \frac{I_1 - I_3}{I_1} \omega_3 \right) \vec{k}'$$

$$= I_3 \omega_3 \tan \theta \left[\sin \left(\frac{I_1 - I_3}{I_1} \omega_3 t + \psi_0 \right) \vec{i}' + \cos \left(\frac{I_1 - I_3}{I_1} \omega_3 t + \psi_0 \right) \vec{j}' \right] + I_3 \omega_3 \vec{k}'$$

证法二

角速度 $\Omega = \frac{I_2 - I_3}{I_1} \omega_3$, 角 $\arctan(\frac{I_3 \omega_3 \tan \theta}{I_3 \omega_3}) = \theta$

2. 取空间坐标系 $Oxyz$ 的角动量 L , 取 $\theta = 0$, $|\dot{\psi}| = \Omega$, $\dot{\psi} = \frac{I_3 \omega_3}{I_1 \cos \theta}$

证法一

3. ω 绕角动量 L 的角速度亦为 $\frac{I_3 \omega_3}{I_1 \cos \theta}$, 夹角 α 满足

$\sin \alpha = \frac{I_3 - I_1}{I_3} \sin \beta \cos \theta$, β 为 ω 与对称轴夹角.

$\vec{\omega} \cdot \vec{L} = (\omega_1, \omega_2, \omega_3) \cdot (I_1 \omega_1, I_2 \omega_2, I_3 \omega_3)$

$= I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2 = C \cos \alpha$

$\omega_1 = A \cos(\frac{I_3 - I_1}{I_1} \omega_3 t + \alpha)$

$\omega_2 = A \sin(\frac{I_3 - I_1}{I_1} \omega_3 t + \alpha)$

$\cos \alpha = \frac{\vec{\omega} \cdot \vec{L}}{|\vec{\omega}| |\vec{L}|} = \frac{C}{(I_1^2 \omega_1^2 + I_2^2 \omega_2^2 + I_3^2 \omega_3^2)^{1/2}} = \frac{C}{(A^2 + \omega_3^2)^{1/2} (I_1^2 A^2 + I_3^2 \omega_3^2)^{1/2}}$

$\sin \alpha = \sqrt{1 - \cos^2 \alpha}$

$= \left\{ \frac{1}{(A^2 + \omega_3^2)(I_1^2 A^2 + I_3^2 \omega_3^2)} \left[I_1^2 A^4 + (I_1^2 + I_3^2) \omega_3^2 A^2 + I_3^2 \omega_3^4 - (I_1 A^2 + I_3 \omega_3^2)^2 \right] \right\}^{1/2}$

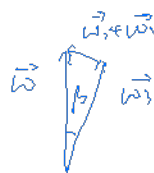
$= \left\{ \frac{1}{(A^2 + \omega_3^2)(I_1^2 A^2 + I_3^2 \omega_3^2)} (I_1 - I_3)^2 A^2 \omega_3^2 \right\}^{1/2}$

$= \frac{(I_1 - I_3) A \omega_3}{L \sqrt{A^2 + \omega_3^2}}$

$= \frac{I_1 - I_3}{I_3} \frac{A}{\sqrt{A^2 + \omega_3^2}} \frac{I_3 \omega_3}{L}$

$= \frac{I_1 - I_3}{I_3} \frac{A}{\sqrt{A^2 + \omega_3^2}} \frac{I_3 \omega_3}{L}$

$= \frac{I_1 - I_3}{I_3} \sin \beta \cos \theta$



另解：坐标

$$\begin{cases} \vec{R} = \vec{i}' \sin \psi \sin \theta + \vec{j}' \cos \psi \sin \theta + \vec{k}' \cos \theta \\ \vec{n} = \vec{i}' \cos \psi - \vec{j}' \sin \psi \\ \vec{k}' = \vec{k} \end{cases}$$

$$\begin{cases} \vec{i}' \sin \psi \sin \theta + \vec{j}' \cos \psi \sin \theta = \vec{R} - \vec{k}' \cos \theta \\ \vec{i}' \cos \psi - \vec{j}' \sin \psi = \vec{n} \end{cases}$$

$$\vec{i}' (\cos \psi + \tan \psi) = \frac{\vec{n}}{\sin \psi} + \frac{\vec{R} - \vec{k}' \cos \theta}{\cos \psi \sin \theta}$$

$$\vec{i}' = \vec{n} \cos \psi + \frac{\vec{R} - \vec{k}' \cos \theta}{\sin \theta} \sin \psi$$

$$\begin{aligned} \vec{\omega} &= \omega_1 \vec{i}' + \omega_2 \vec{j}' + \omega_3 \vec{k}' \\ &= \omega_1 \left(\vec{n} \cos \psi + \frac{\vec{R} - \vec{k}' \cos \theta}{\sin \theta} \sin \psi \right) + \omega_2 \frac{1}{\sin \psi} \left(\frac{\vec{R} - \vec{k}' \cos \theta}{\sin \theta} \sin \psi \cos \psi - \vec{n} \sin^2 \psi \right) + \omega_3 \vec{k}' \\ &= \left(\omega_1 \frac{\sin \psi}{\sin \theta} + \omega_2 \frac{\cos \psi}{\sin \theta} \right) \vec{R} + \left(\omega_1 \cos \psi - \omega_2 \sin \psi \right) \vec{n} \\ &\quad + \left[-\cos \theta (\omega_1 \sin \psi + \omega_2 \cos \psi) + \omega_3 \right] \vec{k}' \\ &= \frac{A}{\sin \theta} \sin \left(\frac{z_3 - z_1}{z_1} \omega_1 t + \alpha + \psi \right) \vec{R} + A \cos \left(\frac{z_3 - z_1}{z_1} \omega_1 t + \alpha + \psi \right) \vec{n} \\ &\quad + \left[-A \cos \theta \sin \left(\frac{z_3 - z_1}{z_1} \omega_1 t + \alpha + \psi \right) + \omega_3 \right] \vec{k}' \end{aligned}$$

$$\vec{L} = \frac{1}{|\vec{r}|} (z_1 \omega_1, z_2 \omega_2, z_3 \omega_3) = (L_k, L_n, L_{k'}) \quad (\vec{R}, \vec{n}, \vec{k}')$$

$$\vec{\omega} \cdot \vec{L}, \quad \vec{\omega} \times \vec{L} \quad \text{---}$$

2. 非对称刚体

$$\begin{cases} I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 \\ I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 \\ I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 \end{cases}$$

角动量守恒: $(I_1 \omega_1)^2 + (I_2 \omega_2)^2 + (I_3 \omega_3)^2 = L^2$

动能守恒: $\frac{1}{2} I_1 \omega_1^2 + \frac{1}{2} I_2 \omega_2^2 + \frac{1}{2} I_3 \omega_3^2 = T$

几何法:

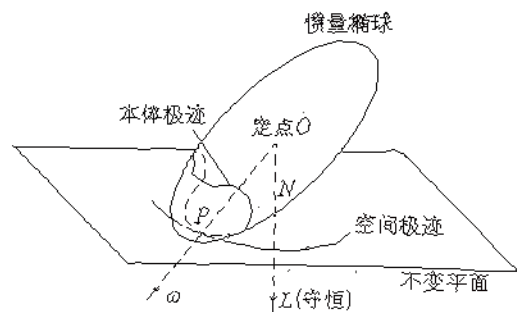
惯量椭球: $I_1 x_1^2 + I_2 x_2^2 + I_3 x_3^2 = 1$

$$\vec{OP} = \frac{1}{\sqrt{2}} \frac{\vec{L}}{\omega}$$

$$\begin{aligned} \vec{ON} &= \vec{OP} \cdot \frac{\vec{L}}{L} = \frac{1}{\sqrt{2} \omega L} \vec{L} \cdot \vec{L} \\ &= \frac{1}{\sqrt{2} \omega L} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) \end{aligned}$$

$$L = I_1 \alpha^2 + I_2 \beta^2 + I_3 \gamma^2 = I_1 \left(\frac{\omega_1}{\omega} \right)^2 + I_2 \left(\frac{\omega_2}{\omega} \right)^2 + I_3 \left(\frac{\omega_3}{\omega} \right)^2$$

$$= \frac{1}{\omega} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) = \frac{\sqrt{2} T}{\omega} \approx \text{const.}$$



3. 动平衡 的稳定说

刚体绕某惯量主轴转动

$$\begin{cases} I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 \\ I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 \\ I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 \end{cases}$$

$$\vec{\omega} = \omega_1 \vec{x}_1 + \omega_2 \vec{x}_2 + \omega_3 \vec{x}_3 \rightarrow \omega_1, \omega_2 \ll 1$$

$$\Rightarrow \dot{\omega}_3 = 0, \omega_3 = \text{const.}$$

$$\begin{cases} \ddot{\omega}_1 + \left[\frac{(I_3 - I_1)(I_3 - I_2)}{I_1 I_2} \omega_3^2 \right] \omega_1 = 0 \\ \ddot{\omega}_2 + \left[\frac{(I_3 - I_1)(I_3 - I_2)}{I_1 I_2} \omega_3^2 \right] \omega_2 = 0 \end{cases}$$

① $I_3 > I_1, I_2$ / $I_3 < I_1, I_2$ 谐振动, 稳定

② $I_1 < I_3 < I_2$ / $I_2 < I_3 < I_1$ 指数增长, 不稳定

* 欧拉动力学方程出发推 $L, T = \text{const.}$

$$\begin{cases} I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 & \textcircled{1} \\ I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 & \textcircled{2} \\ I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 & \textcircled{3} \end{cases}$$

$$\text{d.s. } L^2 = (I_1 \omega_1)^2 + (I_2 \omega_2)^2 + (I_3 \omega_3)^2 = \text{const.}$$

$$I_1^2 \omega_1 \dot{\omega}_1 = \frac{1}{2} \frac{d}{dt} (I_1 \omega_1)^2 = I_1 (I_2 - I_3) \omega_1 \omega_2 \omega_3 \quad \textcircled{1}'$$

$$I_2^2 \omega_2 \dot{\omega}_2 = \frac{1}{2} \frac{d}{dt} (I_2 \omega_2)^2 = I_2 (I_3 - I_1) \omega_2 \omega_3 \omega_1 \quad \textcircled{2}'$$

$$I_3^2 \omega_3 \dot{\omega}_3 = \frac{1}{2} \frac{d}{dt} (I_3 \omega_3)^2 = I_3 (I_1 - I_2) \omega_3 \omega_1 \omega_2 \quad \textcircled{3}'$$

$$\Rightarrow \frac{d}{dt} (L^2) = \left[I_1 (I_2 - I_3) + I_2 (I_3 - I_1) + I_3 (I_1 - I_2) \right] \omega_1 \omega_2 \omega_3 = 0$$

$$\text{d.s. } T = \frac{1}{2} I_1 \omega_1^2 + \frac{1}{2} I_2 \omega_2^2 + \frac{1}{2} I_3 \omega_3^2$$

$$I_1 \omega_1 \dot{\omega}_1 = \frac{d}{dt} \left(\frac{1}{2} I_1 \omega_1^2 \right) = (I_2 - I_3) \omega_2 \omega_3 \omega_1$$

$$I_2 \omega_2 \dot{\omega}_2 = \frac{d}{dt} \left(\frac{1}{2} I_2 \omega_2^2 \right) = (I_3 - I_1) \omega_3 \omega_1 \omega_2$$

$$I_3 \omega_3 \dot{\omega}_3 = \frac{d}{dt} \left(\frac{1}{2} I_3 \omega_3^2 \right) = (I_1 - I_2) \omega_1 \omega_2 \omega_3$$

$$\Rightarrow \frac{d}{dt} T = (I_2 - I_3 + I_3 - I_1 + I_1 - I_2) \omega_1 \omega_2 \omega_3 = 0$$

§ 6.5 对称重刚体的定点运动 (拉格朗日-泊松)

$$Z_1 = Z_2, \quad \vec{M} = mgh \sin \theta \vec{n} \\ = mgh \sin \theta (\cos \psi \vec{e}'_1 - \sin \psi \vec{e}'_2)$$

$$\begin{cases} Z_1 \dot{\omega}_1 = (Z_1 - Z_3) \omega_2 \omega_3 + mgh \sin \theta \cos \psi \\ Z_2 \dot{\omega}_2 = (Z_3 - Z_1) \omega_1 \omega_3 - mgh \sin \theta \sin \psi \\ Z_3 \dot{\omega}_3 = 0 \end{cases} \quad \text{求解};$$

↓ 用 Lagrange 方程

$$L = \frac{1}{2} Z_1 (\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta) + \frac{1}{2} Z_3 (\dot{\psi} + \dot{\varphi} \cos \theta)^2 - mgh \cos \theta$$

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = Z_1 \sin^2 \theta \dot{\varphi} + Z_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \cos \theta = L_2 = \text{const.}$$

$$p_\psi = Z_3 (\dot{\psi} + \dot{\varphi} \cos \theta) = L_3 = \text{const.}$$

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = Z_1 \sin \theta \cos \theta \dot{\varphi}^2 - Z_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \dot{\varphi} \sin \theta + mgh \sin \theta - Z_1 \ddot{\theta} = 0$$

$$\hookrightarrow Z_1 \ddot{\theta} = mgh \sin \theta - (p_\psi - Z_1 \dot{\varphi} \cos \theta) \dot{\varphi} \sin \theta$$

$$H = \sum p_\alpha \dot{q}_\alpha - L = E = \text{const.}$$

$$= \frac{1}{2} Z_1 (\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta) + \frac{1}{2} Z_3 (\dot{\psi} + \dot{\varphi} \cos \theta)^2 + mgh \cos \theta = \text{const.}$$

$$\hookrightarrow \begin{cases} \dot{\psi} + \dot{\varphi} \cos \theta = \frac{L_3}{Z_3} & ① \\ \frac{1}{2} Z_1 (\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta) + \frac{1}{2} \frac{L_3^2}{Z_3} + mgh \cos \theta = E & ② \\ Z_1 \dot{\varphi} \sin^2 \theta + L_3 \cos \theta = L_2 & ③ \end{cases}$$

$$\text{解: } ②, ③ \rightarrow \frac{1}{2} Z_1 \dot{\theta}^2 + \frac{1}{2} \frac{(L_2 - L_3 \cos \theta)^2}{Z_1 \sin^2 \theta} + mgh \cos \theta + \frac{1}{2} \frac{L_3^2}{Z_3} = E$$

$$u = \cos \theta, \quad \dot{u} = -\sin \theta \cdot \dot{\theta}$$

$$\dot{u}^2 = \left(\frac{2L}{I_1} - \frac{L_3^2}{2I_1 I_2} - \frac{2mgh}{I_1} \right) (1-u^2) - \left(\frac{L_2 - L_3 u}{I_1} \right)^2$$

$$\hookrightarrow u = u(t) \dots$$

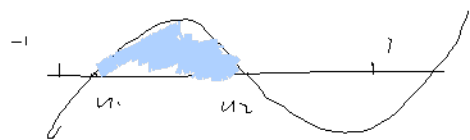
$$\dot{\varphi}(t) = \frac{L_2 - L_3 u}{I_1 (1-u^2)}, \quad \varphi = \int \dot{\varphi} dt = \int \frac{L_2 - L_3 u}{I_1 (1-u^2)}$$

$$\psi(t) = \frac{L_3}{L_2} - \varphi u$$

讨论: $\left(\frac{2L}{I_1} - \frac{L_3^2}{2I_1 I_2} - \frac{2mgh}{I_1} \right) (1-u^2) - \left(\frac{L_2 - L_3 u}{I_1} \right)^2 = \dot{u}^2 \geq 0$

在 $u < -1$ 无零点, 在 $u > 1$ 有一零点 $\Rightarrow$ 在 $[-1, 1]$ 有两零点 u_1, u_2

$\Rightarrow$ 章动: $\theta \in [\theta_1, \theta_2]$, 进动 $\dot{\varphi}$



例: 作 6.2) 对称重刚体绕竖直的对称轴作稳定的进动, 在距离支点 d 的地方

对刚体施加冲量 I , 求证此后最大章动角为 $2\arctan(\frac{Id}{I_3 \omega_3})$

解: $L = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\varphi}^2 \sin^2 \theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\varphi} \cos \theta)^2 - mgh \cos \theta$

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = I_1 \dot{\varphi} \sin^2 \theta + I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \cos \theta = L_2 \quad \bar{\varphi} = \frac{L_2 - L_3 \cos \theta}{I_1 \sin^2 \theta}$$

$$P_\psi = \frac{\partial L}{\partial \dot{\psi}} = I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) = L_3$$

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = 2 I_1 \dot{\varphi}^2 \sin \theta \cos \theta + I_3 (\dot{\psi} + \dot{\varphi} \cos \theta) \dot{\varphi} [-\sin \theta] + mgh \sin \theta - I_1 \ddot{\theta} = 0$$

$$= 2 I_1 \left| \frac{L_2 - L_3 \cos \theta}{I_1 \sin^2 \theta} \right|^2 \sin \theta \cos \theta - L_3 \frac{L_2 - L_3 \cos \theta}{I_1 \sin^2 \theta} \sin \theta + mgh \sin \theta - I_1 \ddot{\theta}$$

$$\theta|_{t=0} = 0, \quad L_2 = L_3, \quad 1 - \cos \theta \approx \frac{\theta^2}{2}, \quad \sin \theta \approx \theta$$

$$= 2 I_1 L_3^2 \frac{1}{I_1^2} \frac{1}{\theta^3} \left(\frac{\theta^2}{2} \right)^2 \left(1 - \frac{\theta^2}{2} \right) - \frac{L_3^2}{I_1} \frac{1}{\theta} \frac{\theta^2}{2} + mgh \theta - I_1 \ddot{\theta}$$

$$\approx \frac{L_3^2}{I_1} \frac{1}{4} \theta - \frac{L_3^2}{I_1} \frac{1}{2} \theta + mgh \theta - I_1 \ddot{\theta}$$

$$\Rightarrow I_1 \ddot{\theta} = -\frac{1}{4} \frac{L_3^2}{I_1} \theta + mgh \theta \approx -\frac{1}{4} \frac{I_3^2}{I_1} \omega^2 \theta$$

$$\theta = A \sin \left(\frac{I_3}{2 I_1} \omega_3 t \right)$$

初始条件: $I_1 \dot{\theta}|_{t=0} = I_1 \cdot \frac{I_3}{2 I_1} \omega_3 A = I_1 \omega_3 A \Rightarrow A = \frac{2 I_1 \omega_3}{I_3 \omega_3}$